

Lab2

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Exercise 1

Question

How do you decide what is the best action to make at each instant? Moreover, if you have two policies, how do you select which one is better? The solution to each of these problems can be different or not.

Answer

To decide the best action to make each instant, we use the Bellman optimal equation. At each state s , we pick the action a that minimizes the expected immediate cost plus the discounted future cost:

$$a^* = \arg \min_{a \in \{AC, AE, AS\}} [C(s, a) + \gamma \sum P(s' | s, a) V^*(s')]$$

To decide which of two policies π_1 and π_2 is better, you compare their value functions. Policy π_1 is at least as good as π_2 if $V^{\pi_1}(s) \leq V^{\pi_2}(s)$ for all states s .

Exercise 2

Question

Implement the value iteration (VI) and policy iteration (PI) algorithms. Show that the policies obtained by both methods display similar cost-to-go values. Which one converges faster? Why? What are the advantages and disadvantages of each one?

Suggestion: Start by modeling the cost function $C(s, a)$. Additionally, track the learning. This can be the number of iterations, the Bellman Error: $\|V_{k+1} - V_k\|_\infty$, or the number of states where the optimal policy changed.

Answer

After implementing the value iteration and policy iteration algorithms, we conclude that they produce the same policy with nearly identical value functions ($9.9\text{e-}5$), which confirms correctness.

PI converges in 2 iterations and VI converges in 1994 iterations, which line up with expectations, as each iteration of PI performs an exact policy evaluation by solving the following linear problem:

$$(I - \gamma P_\pi)V = C_\pi$$

It means that the value function is computed perfectly for the current policy before improving it. On the other hand, VI applies the Bellman operator repeatedly, resulting in a slow linear decay (it only moves values slightly closer to optimal each time). We can clearly see the difference in the following plot:

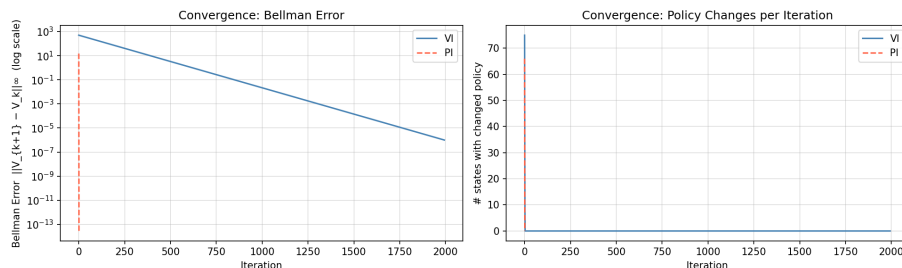


Figure 1: Convergence comparison between Value Iteration and Policy Iteration. Left: Bellman error per iteration (log scale). Right: number of states with changed policy per iteration.

When comparing both algorithms, PI has a bigger cost per iteration than VI, but it's more adequate in small/medium MDP's. Value Iteration performs best in large state spaces where linear solve is prohibitive, but both guarantee convergence.

Exercise 3

Question

What is the impact of the cost?

Suggestion: Run the value iteration with different cost values and see if the convergence rate, policy, mean-time between error changes. Additionally, give a high-level description of all cost functions implemented and add to the answer the following entries from every cost function: $C(0, :)$, $C(5, :)$, $C(32, :)$, $C(49, :)$, $C(75, :)$.

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Answer

We ran the value iteration with these different cost functions: Baseline (original cost function from the assignment) Grid Cheap (reduced cost of buying grid energy) Grid Expensive (increased cost of buying grid energy) Low Failure (reduced failure penalty) High Failure (increased failure penalty) High Solar (increased solar revenue) They look like this:

Config	$C(0,:)$	$C(5,:)$	$C(32,:)$	$C(49,:)$	$C(75,:)$
Baseline	[10, 0, -5]	[10, 0, -5]	[10, 0, -5]	[10, 0, -5]	[500, 500, 500]
Cheap Grid	[5, 0, -5]	[5, 0, -5]	[5, 0, -5]	[5, 0, -5]	[500, 500, 500]
Expensive Grid	[50, 0, -5]	[50, 0, -5]	[50, 0, -5]	[50, 0, -5]	[500, 500, 500]
Low Failure	[10, 0, -5]	[10, 0, -5]	[10, 0, -5]	[10, 0, -5]	[100, 100, 100]
High Failure	[10, 0, -5]	[10, 0, -5]	[10, 0, -5]	[10, 0, -5]	[2000, 2000, 2000]
High Solar	[10, 0, -20]	[10, 0, -20]	[10, 0, -20]	[10, 0, -20]	[500, 500, 500]

Table 1: Cost function entries $C(s, [AC, AS, AE])$ for each configuration across selected states.

From it, we have concluded the only cost value that affect convergence rate is failure penalty, as every other cost value converges in the exact number of iterations (1994) and high/low failure penalty result in 1834/2132 respectively. This small variation occurs because a larger failure penalty creates larger value differences between states, slightly affecting how quickly the Bellman error contracts.

Regarding the mean time between policy changes, all configurations showed a value of approximately 1 iteration, meaning the policy was updated at nearly every iteration throughout VI. The only exception was $\gamma = 0.999$, which showed a mean of 1.67 iterations, reflecting the fact that at very high discount factors the value function changes so slowly that the policy occasionally stays fixed for multiple iterations before updating.

Finally, Table 2 shows the optimal policy across configurations for the selected states. The policy remained identical across all cost configurations, suggesting the relative ordering of actions is robust to cost scaling at these states.

State	Baseline	Cheap Grid	Expensive Grid	Low Failure	High Failure	High Solar
0	AC	AC	AC	AC	AC	AC
5	AS	AS	AS	AS	AS	AS
32	AC	AC	AC	AC	AC	AC
49	AE	AE	AE	AE	AE	AE
75	AC	AC	AC	AC	AC	AC

Table 2: Optimal policy $\pi(s)$ for selected states across cost configurations.

Exercise 4

Question

What is the importance of having the weather information in the state? Will it change the results? Why?

Suggestion: Create a new transition matrix without the weather information, i.e. marginalized over those factors.

Answer

Weather information in the state is critically important and yes, removing it dramatically changes the results. Here’s why:

1. Weather drives the key decision: Solar (AS) vs. Grid Charge (AC) In the full MDP, the optimal policy for 9 out of 15 (B,C) combinations changes depending on the weather. The pattern is very clear from the output:

When $W = 0$ (worst weather): the agent always picks AC (grid charge) because solar charging has only a 5% success rate — it’s essentially gambling. When $W \geq 1$ (any decent weather): the agent switches to AS (solar) because the success probability jumps to 20–95%, making free solar energy the better bet.

Without weather, the marginalized MDP can’t distinguish these situations. It sees only the average solar success rate (61%), which leads it to always default to AC (grid charge) in nearly every non-export state — missing the opportunity to exploit sunny weather for free energy.

2. The cost impact is enormous The value function differences tell the story: the marginalized MDP estimates costs around 780–808 per state, while the full MDP’s weather-averaged values are only 66–77. That’s roughly a 10× higher expected cost when weather is ignored. This happens because the marginalized agent can’t adapt its strategy to current conditions — it pays for grid power even when solar would work well.